

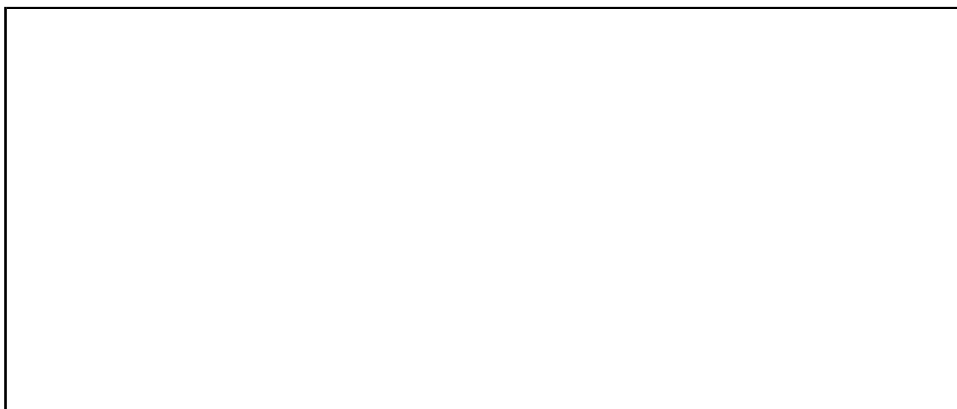
CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

First Semester of M.Sc.(Physics) Examination March 2018

PS 711 Mathematical Methods in Physics-I

Date: 26/03/2018, Monday Time: 10:00 a.m. to 10:30 a.m. Maximum Marks: 10

MCQ



Important Instructions:

- Attempt all questions.
 - Answer the questions in the question paper itself.
 - Use of cell phones is strictly prohibited.
 - Use of non-programmable calculators may be allowed if needed.
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Q.1. Choose the correct answer for the following questions. [10]

1. The area of a parallelogram whose adjacent sides are $\hat{i} - 2\hat{j} + 3\hat{k}$ and $2\hat{i} + \hat{j} - 4\hat{k}$ is _____.

- a. $5\sqrt{6}$ b. $\sqrt{6}$ c. 12 d. -12

2. The maximum rate of change of scalar function ϕ is _____.

- a. $\nabla \cdot \phi$ c. $\nabla \times \phi$
b. $\nabla \phi$ d. none of these.

3. The rank of the matrix $\begin{bmatrix} 1 & 2 & 3 \\ 0 & 7 & 0 \\ 2 & 4 & 6 \end{bmatrix}$ is _____.
- a. 0 b. 1 c. 2 d. 3.
4. The value of the determinant $\begin{vmatrix} 1 & 3 & 4 \\ 0 & 5 & 7 \\ 2 & 8 & 5 \end{vmatrix}$ is _____.
- a. 29 b. -29 c. 42 d. 71.
5. If eigen values of a matrix A are $2, -2$ and 7 , then the determinant of A is _____.
- a. 7 b. -7 c. 28 d. -28
6. A matrix A is called orthogonal, if _____.
- a. $AA^T = A^T A = I$ c. $A^T = A$
b. $A(\bar{A})^T = (\bar{A})^T A = I$ d. $A^T = -A$,

where A^T is the transpose of A and $\bar{A}$ is the conjugate of the matrix A .

7. Which of the following differential equation is linear?
- a. $\frac{dy}{dx} - y = xy^2$ c. $x\frac{dy}{dx} - x^2y^3 = e^x$
b. $y\frac{dy}{dx} - x = \sin x$ d. $\frac{dy}{dx} - xy = \log x$.
8. The wronskian of two functions $\cos 2x$ and $\sin 2x$ is _____.
- a. 1 b. 2 c. $\tan 2x$ d. $\cot 2x$
9. What is the kernel of Fourier transform?
- a. e^{sx} b. e^{-sx} c. e^{isx} d. e^{-isx}
10. Let $f(x) = x^{999}$ defined on $(-5, 5)$. Then Fourier series of $f(x)$ contains _____.
- a. only sine terms c. only constant terms
b. only cosine terms d. none of above

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Date: 26/03/2018, Monday Time: 10:30 a.m. to 01:00 p.m. Maximum Marks: 25

Instructions:

1. Section I and II must be attempted in same answer sheet.
 2. Make suitable assumptions and draw neat figures wherever required.
 3. Use of non-programmable calculator is allowed.
 4. Show necessary calculations.
 5. Figures to right indicate marks.
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SECTION-I

Q-2(a) Find the eigenvalues of the matrix $\begin{bmatrix} 11 & -1 & 5 \\ 0 & 7 & 3 \\ 0 & 0 & -23 \end{bmatrix}$. [02]

(b) Solve the differential equation $\frac{d^2y}{dt^2} - 7\frac{dy}{dt} + 12y = 0$. [02]

(c) Find the Fourier sine transform of $f(x) = \begin{cases} x, & \text{for } 0 < x < \frac{1}{2}, \\ 1 - x, & \text{for } \frac{1}{2} < x < 1, \\ 0, & \text{for } x > 1. \end{cases}$ [03]

OR

(c) Obtain the half range cosine series of x^2 in the interval $(0, \pi)$. [03]

(d) Evaluate $\oint_C \vec{F} \cdot d\vec{r}$, where $\vec{F} = x^2\hat{i} + xy\hat{j}$ and C is the boundary of square in the plane $z = 0$ and bounded by lines $x = 0, y = 0, x = 5$ and $y = 5$. [03]

OR

(d) Find the directional derivative of the function $\psi(x, y, z) = xy^2 + yz^3$ at the point $(1, -1, 1)$ in the direction of $(3, 1, -1)$. [03]

SECTION II

Q-3(a) Solve the differential equation $\frac{dy}{dx} - \frac{y}{x} = x^2 + 1$. [03]

- (b) Find the value of n for which the vector $r^n \bar{r}$ is solenoidal, where $\bar{r} = x\hat{i} + y\hat{j} + z\hat{k}$ [04]
and $r = |\bar{r}|$.

OR

- (b) Using Stoke's theorem, evaluate $\oint_C [(2x - y) dx - yz^2 dy - y^2z dz]$, where C is the [04]
circle $x^2 + y^2 = 4$, corresponding to the surface of sphere of radius 2.

- (c) Test the consistency of the following system of equations and, if consistent, [04]
solve them.

$$2x + 3y + 4z = 11, \quad x + 5y + 7z = 15, \quad 3x + 11y + 13z = 25.$$

OR

- (c) Find the inverse of the matrix $\begin{bmatrix} 0 & 1 & 2 \\ 1 & 2 & 3 \\ 2 & 1 & 1 \end{bmatrix}$, if exist. [04]

- (d) Find a Fourier series to represent $|x|$ in the interval $(-2, 2)$. [04]

OR

- (d) Find the Fourier transform of the function $f(t) = \begin{cases} 1, & \text{for } -2 < t < -1, \\ 2, & \text{for } -1 < t < 1, \\ 1, & \text{for } 1 < t < 2. \end{cases}$ [04]

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